

Predictive Maintenance Approach

Brief A of 2 | Prepared by Valentine Kerubo | Week 6

SCIENCE: MAINTENANCE / RELIABILITY ENGINEERING

This brief documents the sensor logic, feature engineering, and model performance behind the pump failure-prediction system, built on 28 days of 6-hourly readings across a 24-pump fleet.

1. Sensor Inputs & Feature Logic

Five raw channels feed the model per reading: vibration (g), temperature (°C), pressure (psi), acoustic level (dB), and motor current (A), plus asset age and cumulative error events. Because a single reading is noisy, we derive four rolling features per asset over a 4-reading (1-day) window:

Feature	Logic	Why it matters
vib_roll_mean	Rolling mean of vibration_g	Smooths sensor noise to expose trend
vib_rms	Rolling RMS of vibration_g	Standard vibration-analysis severity metric; sensitive to spikes
vib_range	Rolling (max – min) of vibration_g	Rising range signals bearing-wear instability
temp_roll_std	Rolling std. dev. of temperature_c	Captures thermal instability preceding seal/bearing failure

Sensor dropouts in pressure_psi / acoustic_db (<1% of readings) are imputed with each asset's own median rather than dropped, to preserve the time ordering rolling features depend on.

2. Model & Validation Design

A Logistic Regression classifier (class-weight balanced, standardized features) predicts **failure_within_7_days** from the 11 raw + engineered features above. Critically, the train/test split is by **asset_id**, not by row (18 pumps train / 6 pumps held out entirely), a random row-level split would leak a pump's own baseline into the test set and overstate performance. This mirrors the real deployment case: scoring pumps the model has never seen.

3. Performance on Held-Out Pumps (n = 660 readings, 166 true failures)

Metric	Value	Engineering read
Sensitivity (Recall)	83.7%	139 / 166 true failures flagged in advance
Specificity	95.5%	472 / 494 healthy readings correctly cleared
False Positive Rate	4.5%	22 healthy readings incorrectly flagged
Precision	86.3%	139 / 161 flagged readings were real failures
Accuracy	92.6%	Overall correct classification rate
ROC AUC	0.92	Strong separation between classes

4. Known Limitations

- The 27 false negatives (16.3% of true failures) are the main operational risk, class-weighting was tuned toward recall, but some failures still slip through and existing inspection routines should remain in place as a backstop.
- A companion Linear Regression model estimating hours-to-failure explains only ~46% of variance (MAE ≈ 118h), so treat its output as a coarse prioritization signal, not a scheduling deadline.
- Model is trained on 28 days of synthetic data across 24 assets; recommend re-validating thresholds quarterly as real production data accumulates.

Smarter Pump Maintenance: The Business Case

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AUDIENCE: CFO / FINANCE LEADERSHIP

We built a system that watches our pump fleet around the clock and warns us, days in advance, when a pump is heading toward a breakdown, so we can fix it on our schedule, not the machine's.

The Problem Today

When a pump fails without warning, we pay for it twice: once in the emergency repair (rush parts, overtime labor, expedited shipping), and again in the production time lost while it's down. Unplanned breakdowns are simply the most expensive way to maintain equipment.

What Changes With This System

Out of every 100 pumps genuinely at risk of failing within a week, the system correctly catches about 84 of them in advance, giving the maintenance team a heads-up to schedule a repair during planned downtime instead of scrambling after the fact. It also stays quiet: it only raises a false alarm on roughly 4 or 5 out of every 100 healthy pumps, so it doesn't create a habit of ignored warnings.

Estimated Financial Impact (illustrative, fleet of 24 pumps)

	Amount
Average cost of one unplanned breakdown (emergency repair + downtime)	\$18,000
Average cost of the same repair, done proactively on a planned schedule	\$4,000
Savings per breakdown we catch in advance	\$14,000
Breakdowns avoided per year (est.)	~20 of 24
Estimated gross annual savings	\$280,000
Cost of false-alarm inspections (est.)	-\$20,000
Estimated annual cost to run the system	-\$30,000
Estimated net annual benefit	\$230,000
One-time setup investment	\$150,000
Estimated payback period	< 8 months
Estimated first-year ROI	~150%

Figures above are planning-level estimates based on typical industrial repair/downtime cost ratios; we recommend validating against our own maintenance ledger before finalizing a business case.

Beyond the Dollars: Risk Reduction

- **Fewer surprises.** Shifts spending from reactive firefighting to planned maintenance windows we control.
- **Safety & compliance.** Catching a pump before it fails reduces the chance of a safety incident or environmental release tied to a sudden equipment failure.
- **Better use of the maintenance budget.** Technicians spend time on the ~20% of pumps that need attention, not walking the full fleet on a fixed calendar.

Bottom Line

For a one-time investment of roughly \$150,000, we estimate this system pays for itself in well under a year and then continues returning over \$200,000 annually in avoided breakdown costs, while also lowering our exposure to safety and production risk.

Reflection: Switching Between Audiences

Switching from Brief A to Brief B was harder than expected, not because the underlying facts changed, but because the unit of meaning had to change. In the engineering brief, a number like "83.7% sensitivity" is the finding, it stands on its own and an engineer can immediately judge whether that's good enough. For the executive brief, that same number is only useful once it's translated into something a CFO budgets against: dollars avoided, payback period, risk exposure. The hardest part wasn't hiding the math, it was resisting the instinct to explain *how* the model works before saying *why it matters*, an engineer wants the mechanism first and the payoff second, while an executive wants the payoff first and will only ask about mechanism if the payoff is compelling. Once I made that ordering the organizing principle for each page, rather than just swapping vocabulary, the rest of each brief followed fairly naturally. The one place I had to be deliberate was honesty under simplification: the executive brief still needed to say plainly that ~16% of failures are still missed and that the cost figures are estimates, just without the confusion-matrix language used to say the same thing in Brief A.